

Care of the Critically Ill Patient

The Big Picture

Critical care = structured, team-based management of life-threatening illness. The backbone is the **ATLS approach to trauma**: four phases — **Primary survey, Resuscitation, Secondary survey, Definitive care** — with the first two done **simultaneously**. The mantra is **ABCDE**, find-and-fix the immediately lethal injuries before moving on.

The Four Phases

1. **Primary survey (ABCDE)** — find immediately life-threatening injuries.
2. **Resuscitation** — done with the primary survey.
3. **Secondary survey** — head-to-toe, once stable.
4. **Definitive care** (+ tertiary survey).

A — Airway + Cervical Spine

- Priority: clear airway + oxygenation. Basic manoeuvres ± oro/nasopharyngeal airway; **high-flow O₂ via mask + reservoir**.
- **Assume spinal injury** in significant blunt trauma until cleared (exam unreliable if head injury, intoxication, distracting pain, ↓ consciousness).
- **Intubation indications**: airway obstruction, impending obstruction (burns/inhalation), **GCS <9**, maxillofacial haemorrhage, respiratory failure, need for resuscitative surgery, uncooperative needing investigation.
- Technique: **manual in-line stabilisation** (no traction), preoxygenate, IV induction (**ketamine** least hypotensive if hypovolaemic — give fluid first), short-acting relaxant, **cricoid pressure**, bougie + video-laryngoscope. Can't intubate → supraglottic (i-Gel) → **surgical cricothyroidotomy** (6.0 ETT).

B — Breathing (immediately life-threatening chest injuries)

- **Tension pneumothorax** — reduced movement/breath sounds, hyperresonance, distress + hypotension + tachycardia (tracheal deviation = late). **Immediate decompression**: cannula 2nd ICS midclavicular **or** thoracostomy 5th ICS anterior axillary → then chest drain (32 FG).
- **Open pneumothorax** — occlusive dressing **sealed on 3 sides** (flutter valve) + chest drain away from wound.
- **Massive haemothorax** — **>1500 mL**; dull percussion + hypovolaemia. Fluids + chest drain; **thoracotomy if drain loss >200 mL/h**.
- **Cardiac tamponade** — distended neck veins + hypotension (penetrating injury); **FAST** diagnoses; deteriorating → thoracotomy + pericardiectomy (pericardiocentesis temporises).

C — Circulation (hypovolaemia)

- Control external bleeding with direct pressure. **Four classes of hypovolaemic shock** by % blood loss.
- ↓ Systolic BP suggests **>30% loss (~1500 mL)**; ↓ consciousness ⇒ **40–50% loss**.
- **Two large-bore (14G) IV cannulae**; FBC + cross-match. Difficult access → external jugular/femoral or **intraosseous** (200 mL/min). Arterial line for BP + ABG (base deficit guides resus).

- **Permissive hypotension** — before bleeding controlled, aggressive fluids are **harmful** (dislodge clot, dilute clotting). Give **250 mL crystalloid boluses to keep systolic ~80 mmHg** (higher if older/head injury).
- **Blood early** — platelets/FFP/red cells ~**1:1:1**; **tranexamic acid 1 g within 3 h**; group O immediately, group-specific in 10 min, full cross-match 45 min.
- **Warm all fluids; avoid hypothermia** (<35°C = independent mortality predictor — impairs coagulation, causes arrhythmia, ↑ infection).

D — Disability

Pupils + **GCS** (do a quick neuro check before intubating).

E — Exposure

Undress fully; **prevent hypothermia** with warm blankets.

Tubes & Imaging

- **Urinary catheter** (output = resus indicator), **gastric tube** (orogastric if basal skull fracture suspected).
- **Whole-body CT** primary investigation, within 30 min; chest/pelvic X-ray if going straight to theatre or very unstable. C-spine assumed injured until CT clears.

Secondary Survey

Head-to-toe once vitals stable; **re-evaluate repeatedly** for ongoing bleeding; exsanguinating haemorrhage may need laparotomy as part of resuscitation.

Walk-Away Points

- **Primary + resuscitation happen together**; ABCDE order.
- **Tension pneumothorax = clinical diagnosis** → decompress before X-ray.
- **Permissive hypotension** + early blood (1:1:1) + **TXA within 3 h**; avoid crystalloid flooding.
- **Hypothermia kills** — warm fluids, cover the patient.
- Whole-body CT in 30 min; output + falling lactate confirm adequate resuscitation.

Care of the Critically Ill Patient — Revision Table	
Phase	Note
Primary survey	ABCDE — find lethal injuries
Resuscitation	Simultaneous with primary
Secondary survey	Head-to-toe when stable
Definitive + tertiary	Final care
A — Airway	Detail
Assume	C-spine injury until cleared
Intubate if	Obstruction, GCS<9 , resp failure
Technique	Manual in-line stabilisation, cricoid, bougie
Can't intubate	i-Gel → surgical cricothyroidotomy
B — Breathing emergency	Treatment
Tension pneumothorax	Decompress 2nd ICS MCL / 5th ICS AAL → drain
Open pneumothorax	3-sided occlusive dressing + drain
Massive haemothorax	>1500mL ; thoracotomy if >200mL/h
Cardiac tamponade	FAST; thoracotomy/pericardiocentesis
C — Circulation	Detail
↓Systolic BP	>30% loss (~1500mL)
↓Consciousness	40–50% loss
Access	2× 14G IV / intraosseous
Permissive hypotension	250mL boluses, systolic ~80mmHg
Blood	1:1:1 RBC:FFP:platelet
TXA	1g within 3 h
D / E	Detail
Disability	Pupils + GCS
Exposure	Undress, prevent hypothermia
Hypothermia	<35°C = mortality predictor
Tubes / imaging	Detail
Catheter	Output = resus marker
Gastric tube	Orogastric if basal skull #
Whole-body CT	Within 30 min

Bold = high-yield.